Scam Variant Observatory

Tracking scam mutations in Sri Lanka

ONE-LINE PROBLEM STATEMENT

Scam variant surveillance for Sri Lanka: detecting emerging patterns in Sinhala, Tamil, English, Singlish, and Tanglish scam messages, tracking changes in impersonation tactics, and issuing evidence-based alerts for analyst review.

Problem and users. Scammers continually change their wording, language, impersonated institutions, and payment requests. A detector can become unreliable without operators noticing. Sri Lankan fraud analysts, community digital-safety organisations, and citizens need a way to recognise recurring new patterns and respond when established detection methods become less dependable.

The idea. Inspired by public-health monitoring of virus variants, the system treats scam mechanisms as **families** and changes in their presentation as **variants**. It groups related suspicious reports and alerts an analyst when an unfamiliar pattern recurs. This analogy describes message patterns; it does not establish attacker identity or actual ancestry.

Six-hour prototype. Multilingual embeddings and a trained classifier detect known scams. Similarity and novelty scores group unfamiliar messages. A dashboard shows prediction confidence, unfamiliar-message rates, and emerging clusters. Candidate-variant alerts enter an analyst queue; confirmed examples support a controlled model update followed by re-evaluation.

Adversarial stress testing. An offline LLM-based adversarial agent generates fictional variants by changing wording, language, impersonation targets, or requested actions. A clearly labelled hypothetical SL-UDI impersonation scenario demonstrates a new institutional target. These simulations test resilience; they do not forecast future scams.

Data and evaluation. During the build window, we will combine appropriately licensed public phishing data with generated, human-reviewed Sri Lankan examples. Language coverage will be evaluated separately and limited to what the team can validate. Training, validation, held-out test, and shifted sets will be separate, with related translations and paraphrases kept together. Measures include precision, recall, unseen-family detection, false alerts on unfamiliar legitimate messages, reports before an alert, and review workload. Post-update evaluation will use examples excluded from retraining.

Delivery and impact. Deliverables include a runnable application, reproducible ML pipeline, prediction logging, demonstrated drift detection and response, documented results, an impact statement, and a PDF pitch deck. No real personal identifiers will be used; synthetic scams remain restricted to training and testing. The intended benefit is earlier analyst recognition and escalation of changing scam patterns, demonstrated through a controlled simulation.